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PII: S1872-4973(17)30092-3
DOI: http://dx.doi.org/doi:10.1016/j.fsigen.2017.04.008
Reference: FSIGEN 1704

To appear in: *Forensic Science International: Genetics*

Received date: 10-1-2017
Revised date: 31-3-2017
Accepted date: 13-4-2017

Please cite this article as: Corina C.G.Benschop, Linda van de Merwe, Jeroen de Jong, Vanessa Vanvooren, Morgane Kempenaers, C.P.(Kees) van der Beek, Filippo Barni, Eusebio López Reyes, Léa Moulin, Laurent Pene, Hinda Haned, Titia Sijen, Validation of SmartRank: A likelihood ratio software for searching national DNA databases with complex DNA profiles, Forensic Science International: Geneticshttp://dx.doi.org/10.1016/j.fsigen.2017.04.008

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Validation of SmartRank: A likelihood ratio software for searching national DNA databases with complex DNA profiles

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Highlights

- The SmartRank software was developed, extensively tested and validated
- The methodology enables searching voluminous DNA databases with complex profiles
- Applying SmartRank may provide investigative leads to criminal cases lacking a suspect
- SmartRank goes beyond CODIS in complex profile searches
- We provide guidelines when and how to use SmartRank for best results

Abstract

Searching a national DNA database with complex and incomplete profiles usually yields very large numbers of possible matches that can present many candidate suspects to be further investigated by the forensic scientist and/or police. Current practice in most forensic laboratories consists of ordering these 'hits' based on the number of matching alleles with the searched profile. Thus, candidate profiles that share the same number of matching alleles are not differentiated and due to the lack of other ranking criteria for the candidate list it may be difficult to discern a true match from the false positives or notice that all candidates are in fact false positives. SmartRank was developed to put forward only relevant candidates and rank them accordingly. The SmartRank software computes a

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likelihood ratio (LR) for the searched profile and each profile in the DNA database and ranks database entries above a defined LR threshold according to the calculated LR. In this study, we examined for mixed DNA profiles of variable complexity whether the true donors are retrieved, what the number of false positives above an LR threshold is and the ranking position of the true donors. Using 343 mixed DNA profiles over 750 SmartRank searches were performed. In addition, the performance of SmartRank and CODIS were compared regarding DNA database searches and SmartRank was found complementary to CODIS. We also describe the applicable domain of SmartRank and provide guidelines. The SmartRank software is open-source and freely available. Using the best practice guidelines, SmartRank enables obtaining investigative leads in criminal cases lacking a suspect.

Keywords Forensic Science; DNA database; CODIS; Likelihood Ratio; LRmix; SmartRank

1. Introduction

DNA profiles that are both incomplete and mixed are commonly encountered in forensic casework. Searching a DNA database with such profiles usually yields very large numbers of possible matches representing many candidate suspects to be further investigated by the forensic scientist and/or police. Current practice in most forensic laboratories consists of grouping these 'hits' based on the number of matching alleles and manually compare the candidates in the most promising groups to the evidentiary trace profile while considering the rarity of the alleles and peak heights. Candidates that share the same number of matching alleles are not differentiated and a true match may thus be hidden by false positive candidates. For complex profiles, the true contributors may not appear on the candidate list, while false positives may. To address this issue, an approach is needed that ranks the candidates by using additional information in the profiles and a likelihood ratio (LR) score has been demonstrated to better represent the strength of association between the candidates in the database and the searched profile [1,2,3,4]. Within the LR framework, all individuals in the DNA database are evaluated as a potential donor and a LR is calculated. Individuals whose profiles lead to high LRs can be treated as persons of interest by the investigators. LRmix is one such LR model enabling the interpretation of complex DNA profiles [5,6] that may be affected by drop-in, drop-out and comprise a high number of contributors. The LRmix model was modified to enable searching of voluminous DNA databases and translated into the user-friendly software SmartRank [7]. In this study, we subjected the SmartRank software to extensive robustness testing using databases containing anonymized STR entries stored in national DNA databases of varying size and composition, and using DNA mixtures with various characteristics. In addition, various mixed profiles were used in both SmartRank and CODIS searches to infer the profile types for which SmartRank can be

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complementary to CODIS. We used the results to generate guidelines for best practice and define the applicable domain of the SmartRank software.

2. Materials and methods

For the description of the SmartRank software we refer to Ref. [7].

2.1 Selection of multi-locus reference genotypes

A total of 44 reference profiles were used consisting of 15 STR loci (D3S1358, vWA, D16S539, D2S1338, D8S1179, D21S11, D18S51, D19S433, TH01, FGA, D10S1248, D22S1045, D2S441, D1S1656, D12S391). First, 23 profiles were selected from a reference set of 2,085 Dutch males [8,9] and to ten of these, a single locus genotype for locus SE33 was added while taking care that these SE33 alleles were observed more than once in the 2,085 reference set [9]. Next, 21 virtual genotypes were derived from these 23 genotypes to generate a set denoted 'resembling donors': for five resembling genotypes one randomly chosen allele was replaced by a rare allele (having a frequency of 0.00024 in [9]); for 12 resembling genotypes alleles from two or three of the selected 23 genotypes were combined in such a manner that the resembling genotype had 100% (n=4), 75% (n=4) or 50% (n=4) overlap with a two- or three-person mixture of these selected two or three genotypes, and for the remaining four resembling genotypes existing genotypes were adapted to represent a father (n=2; at least one allele per locus in common) or a brother (n=2; sharing half of the overall profile) of one of the 23 donors. This set of 44 reference genotypes forms the basis of the SmartRank performance study: they were added to anonymized copies of DNA databases and used to derive mixed DNA profiles (see section 2.2) to search these databases.

2.2 Simulated DNA mixtures for database searches

Based on the 44 reference profiles described in section 2.1, 343 mixed DNA profiles were created (see Supplementary Table 1) that had two to five contributors (155 two-person, 155 three-person, 16 four-person and 17 five-person mixtures) with variable levels of allele sharing (i.e. the alleles that are identical between and/or within contributors; 6 had much allele sharing, 6 had few allele sharing, 331 had moderate allele sharing), five to 16 loci (19 16-loci, 36 15-loci, 72 8-loci, 72 7-loci, 72 6-loci and 72 5-loci mixtures) in various combinations (see Supplementary Table 2), a variable percentage of drop-out (12 mixtures with 0%, 21 with 5%, 144 with 10%, 3 with 15%, 1 with 20%, 2 with 30%, 144 with 40% and 16 with 50%) and presence or absence of a rare allele (14 mixtures with and 329 without a rare allele) (Supplementary Table 1).

2.3 Size and composition of DNA databases used

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In this study, anonymized copies of (parts of) the national DNA databases of Belgium, The Netherlands, France, Italy and Spain were used that had variable sizes as depicted in Table 1. In addition, a simulated DNA database (Table 1) was used for examining the effects of the drop-out estimation on the LRs which is described in section 2.4. To each of these DNA databases the 44 reference profiles described in section 2.1 were added to examine retrieval of true donors for the simulated DNA mixtures described in section 2.2 upon SmartRank or CODIS (Combined DNA Index System) searches.

Table 1. Size and composition (percentage of profiles with a given number of loci) for the DNA databases that were used in this study. The number of profiles in the DNA databases excludes the Amelogenin locus and includes the 44 reference profiles that were used in this study. Frequencies between 10% and 50% are coloured grey; those of 50% and higher black.

	DNA database and size							
Number of loci	**France (large)**	**France (small)**	**The Netherlands (2015)**	**The Netherlands (2013)** [a]	**Spain**	**Italy**	**Belgium**	**Simulated DNA database** [b]
	***n*=1,552,754**	***n*=43,861**	***n*=206,535**	***n*=175,890**	***n*=189,865**	***n*=69,985**	***n*=37,595**	***n*=220,044**
3	-	-	-	<0.01%	-	0.10%	-	-
4	-	-	-	<0.01%	-	0.12%	-	-
5	-	-	-	<0.01%	<0.01%	0.12%	-	-
6	-	-	<0.01%	0.01%	0.02%	0.28%	<0.01%	-
7	<0.01%	-	<0.01%	<0.01%	0.02%	0.43%	0.02%	-
8	-	-	<0.01%	<0.01%	<0.01%	0.87%	0.05%	0.01%

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9	-	-	0.81%	0.99%	0.02%	1.94%	0.36%	0.05%
10	-	-	51.74%	61.70%	0.39%	6.20%	8.33%	0.42%
11	<0.01%	-	0.05%	0.05%	0.04%	1.13%	2.52%	2.08%
12	<0.01%	-	0.01%	<0.01%	0.21%	1.91%	0.32%	6.41%
13	0.19%	-	0.35%	0.25%	0.15%	3.05%	1.19%	13.59%
14	2.59%	-	3.06%	2.59%	1.17%	2.76%	0.05%	20.45%
15	48.72%	0.07	43.46%	34.03%	66.32%	64.41%	25.14%	22.49%
16	2.07%	0.04%	0.11%	0.09%	5.73%	12.52%	23.93%	17.79%
17	38.74%	0.33%	<0.01%	<0.01%	0.11%	2.97%	22.36%	10.51%
18	0.07%	0.75%	0.01%	<0.01%	0.02%	0.09%	4.18%	4.46%
19	0.17%	1.89%	0.05%	0.02%	0.02%	0.04%	-	1.41%
20	0.25%	2.95%	0.34%	0.22%	0.18%	0.15%	0.01%	0.31%
21	0.40%	5.08%	0.02%	<0.01%	25.05%	0.53%	4.60%	0.04%
22	0.80%	10.58%	<0.01%	<0.01%	0.53%	0.31%	0.66%	<0.01%
23	6.01%	78.32%	-	-	<0.01%	0.07%	6.28%	-

[a] Used in CODIS comparisons only.

[b] Used for examining effects of the applied percentile of the drop-out estimation in SmartRank analyses.

2.4 SmartRank analyses

A total of 770 SmartRank analyses were performed using SmartRank software version 1.0.0: 368 using the Dutch DNA database copy of 2015, 352 using copies of the France (small), Spanish, Italian and Belgian DNA databases (88 searches for each of these four databases), 20 using the France (large) DNA database and 30 using the Dutch DNA database copy of 2013 (Supplementary Tables 3 and 4). These SmartRank analyses were performed within each country holding the indicated DNA databases for which a copy of the SmartRank software was locally installed.

A list with hypotheses and sample files (.csv format) of the mixed profiles was provided (an overview of the hypotheses can be found in Supplementary Table 5). The true number of contributors was used for all hypotheses. Some hypotheses had conditioning on one or two known profiles under both the prosecution and defense hypothesis (Supplementary Tables 3 and 4). In all SmartRank analyses, a probability of drop-in of 0.05 and a theta correction of 0.01 were applied. For the majority of the analyses, a probability of drop-out was given to enter in SmartRank, but for ten analyses, the five countries used the drop-out estimation option in SmartRank. This drop-out estimation follows the Markov Chain Monte Carlo (MCMC) method as implemented in the LRmix model [5]. In case this drop-out estimation option was selected, the LRs were presented that were obtained with the 95% percentile of the estimated probability of drop-out. In order to further investigate the effect of the probability of drop-out, SmartRank software version 1.0.2 was used (lrmixstudio.org/smartrank), with the modification that the applied percentile of the drop-out estimation was made configurable. For the 25 samples in Supplementary Table 1 that are denoted 'Samples used for drop-out estimation comparison', the number of true positives and their ranking was compared when using the 5%, 50% or 95% percentile of this drop-out estimation. In these analyses, a simulated DNA database of size 220,044 was used [7] (Table 1).

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France and The Netherlands used Dutch allele frequencies [9], Italy used Caucasian Italian allele frequencies [10,11] and Spain and Belgium used their own population statistics.

2.5 DNA database searches using CODIS

The partners in the Netherlands, Belgium and Spain performed CODIS (Combined DNA Index System, version 7.0) searches, each using the 30 samples indicated in Supplementary Table 1. These samples were used for both a CODIS search that did not allow any mismatch and a search that allowed one mismatch, which resulted in a total of 180 CODIS search results. For all searches the moderate search stringency was applied. This search stringency results in a match if alleles of one profile (with the least number of alleles) are present in the other profile. To enable inference of profiles types for which SmartRank can complement CODIS, both CODIS and SmartRank searches were performed using the exact same DNA database composition (Table 1, the Netherlands used the copy of 2013).

3. Results and discussion

3.1 Retrieval of true donors versus non-donors for mixtures without drop-out

First, the performance of SmartRank on five different DNA databases was examined with 20 two, three- and four-person mixtures without drop-out. These SmartRank searches resulted in the retrieval (LR >1) of all of the true donors to these mixtures (Fig. 1, green diamonds); *i.e.* the two donors to the two-person mixtures, the three donors to the three-person mixtures, etc. Known donors were not searched for, thus only two donors could be retrieved for three-person mixtures if one known donor was anchored under both hypotheses. Results were irrespective of the DNA database or population statistic that was used. Besides these true donors, resembling donors were retrieved, especially those that were simulated to have 100% overlap with alleles present in the mixed profile (Fig. 1, blue squares). The retrieval of non-donors (a.k.a. false positives or adventitious matches) was dependent on the size and composition of the DNA database. The effect of database size is implied from the observation that the smallest two DNA databases, Belgium and ‘small France’ having sizes of 37,595 and 43,861, respectively, did not yield false positives, while the larger databases. The Netherlands, Spain, Italy and ‘large France’ with sizes of 206,535, 189,867, 69,985 and 1,552,754, respectively, did (Fig. 1, red triangles). An effect of database composition is implied from the observation that more false positives were obtained with the Italian DNA database than with the Spanish database even though it is substantially smaller in size. An (additional) effect of a different population statistic used by each country, however, cannot be ruled out.

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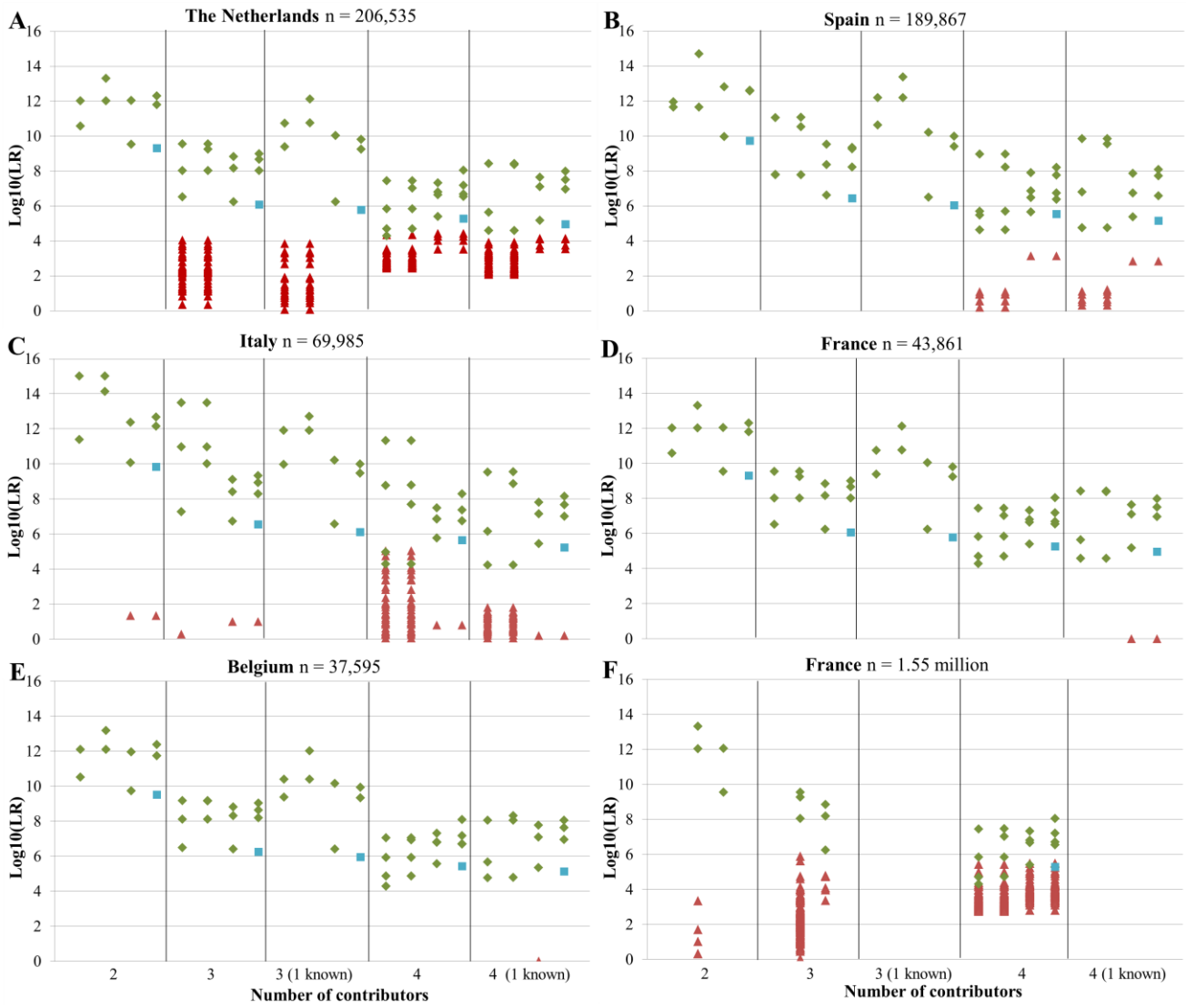


Figure 1. Retrieval of true (green diamonds), resembling (blue squares) and non-donors (red triangles) for two-, three- or four-person mixtures without drop-out. Results obtained when using the Dutch (A) Spanish (B), Italian (C), small France (D), Belgian (E) or large France (F) DNA database. Each vertical line represents the analysis of one mixed profile. Note that only part of the samples were analysed using the large France DNA database (Supplementary Table 3) and known contributors were not searched for.

3.2 Effects of the complexity of the STR profile

In this section, we examine the effects of various mixture characteristics on the retrieval of true donors using the SmartRank software using the various databases that differ in size and composition. Our assumption is that the more complex or less informative a DNA mixture is (*e.g.* more contributors, more drop-out, less loci, more common alleles, high level of allele sharing), the lower the LR for the true donors to the DNA mixture. From Fig. 1 it can be seen that a larger number of contributors indeed inclines to lower the LR for the true donors. It also tends to increase the LR for

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non-donors, resulting in smaller differences between the LRs for true and non-donors. The height of the LR and whether true donors can be distinguished from non-donors also depends on the absolute DNA contribution of a true donor to a DNA mixture, which translates to the percentage of drop-out. With a higher percentage of drop-out, less true donors are retrieved by SmartRank (LR >1) and the ranges in LRs for true donors become less distinguishable, or undistinguishable, from the ranges of LRs for non-donors (Supplementary Fig. 1). When considering the combination of drop-out level and number of contributors, which indicates the absolute and the relative contribution of a true donor, we see the trend that more drop-out may occur for mixtures with less contributors to obtain retrieval of the true donor (Table 2 and Supplementary Fig. 1).

Table 2. Percentages of true donors retrieved by SmartRank for two- to five-person mixtures with varying drop-out rates Note that all profiles have genotyping information on 15 or 16 loci.

		Percentage of true donors retrieved					
Average percentage of drop-out [a]		**0%**	**5%**	**15%**	**20%**	**30%**	**50%**
Two-person mixtures	0 knowns	100%	100%	100%	-	100%	0%
Three-person mixtures	0 knowns	100%	100%	100%	33%	40%	0%
	1 known	100%	-	-	-	-	-
Four-person mixtures	0 knowns	100%	95%	-	-	-	0%
	1 known	100%	-	-	-	-	-
Five-person mixtures	1 known	-	72%	-	-	-	0%
	2 knowns	-	78%	-	-	-	0%
Total number of hypotheses [b]		20	22	3	1	2	22
Total number of SmartRank analyses		100	30	15	5	10	30

[a] Contributors can vary in drop-out percentage and a major contributor may have no drop-out. When a known contributor is used, this is a contributor without drop-out. Known donors are not searched for.
[b] Some SmartRank analyses were performed by all partners using the Belgium, Spain, Italy, small France and Netherlands 2015 databases, for some analyses only the Netherlands 2015 database was applied. Note that the results for mixtures without drop-out are also presented in Figure 1.

Given that forensic DNA profiles may be incomplete due to the compromised state of a sample, we also examined the effect of a lower number of loci in a mixed profile. Two- and three-person mixed profiles with five, six, seven or eight loci, three different loci combinations and 10% drop-out were regarded by all partners. Six additional loci combinations and profiles with 40% drop-out were regarded by The Netherlands (Supplementary Table 1 and Supplementary Table 2). We assessed whether the true donors resided amongst the 250 candidates with the highest LR (all >1), note that in some of the earlier described analyses with 15 or 16 loci, long candidate lists were obtained, especially with the larger databases. Results are presented in Table 3. These analyses show that a lower number of loci negatively influences the retrieval of true donors. When a true donor is not among the top 250 in the ranking list, this donor may still have a LR >1, but in these analyses they are regarded as false negatives irrespective of a LR above or below 1. These false negatives predominantly occurred for the larger DNA databases, three-person mixtures and mixtures missing several alleles (40% drop-out). Large lists of candidates, and thus large numbers of false inclusions,

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compromise the efficiency of a DNA database search, because they infer an enormous effort to investigate all candidates [12]. We consider a list of ten, or possibly 20, candidates as a manageable list. In these analyses with five to eight loci, the true donor resided amongst the top 20 ranking candidates for the majority of two-person mixtures with 10% drop-out, but not for three-person mixtures with 10% drop-out (Supplementary Fig. 2) or mixtures with 40% drop-out (Fig. 2).

Table 3. Percentages of true donors retrieved by SmartRank amongst the top 250 ranking candidates (LR >1) for two- and three-person mixtures having 10% or 40% drop-out and a number of loci varying from five to eight.

Number of loci	DNA database used	Number of SmartRank analyses	10% drop-out		40% drop-out	
			Two-person mixtures	Three-person mixtures	Two-person mixtures	Three-person mixtures
5	The Netherlands (2015)	18	72%	35%	33%	0%
	Spain	6	92%	55%	-	-
	Italy	6	92%	56%	-	-
	France (small)	6	100%	72%	-	-
	Belgium	6	100%	67%	-	-
6	The Netherlands (2015)	18	86%	52%	42%	0%
	Spain	6	100%	56%	-	-
	Italy	6	100%	78%	-	-
	France (small)	6	100%	83%	-	-
	Belgium	6	100%	89%	-	-
7	The Netherlands (2015)	18	94%	72%	39%	0%
	Spain	6	100%	78%	-	-
	Italy	6	100%	89%	-	-
	France (small)	6	100%	94%	-	-
	Belgium	6	100%	94%	-	-
8	The Netherlands (2015)	18	97%	93%	36%	2%
	Spain	6	100%	94%	-	-
	Italy	6	100%	89%	-	-
	France (small)	6	100%	94%	-	-
	Belgium	6	100%	94%	-	-

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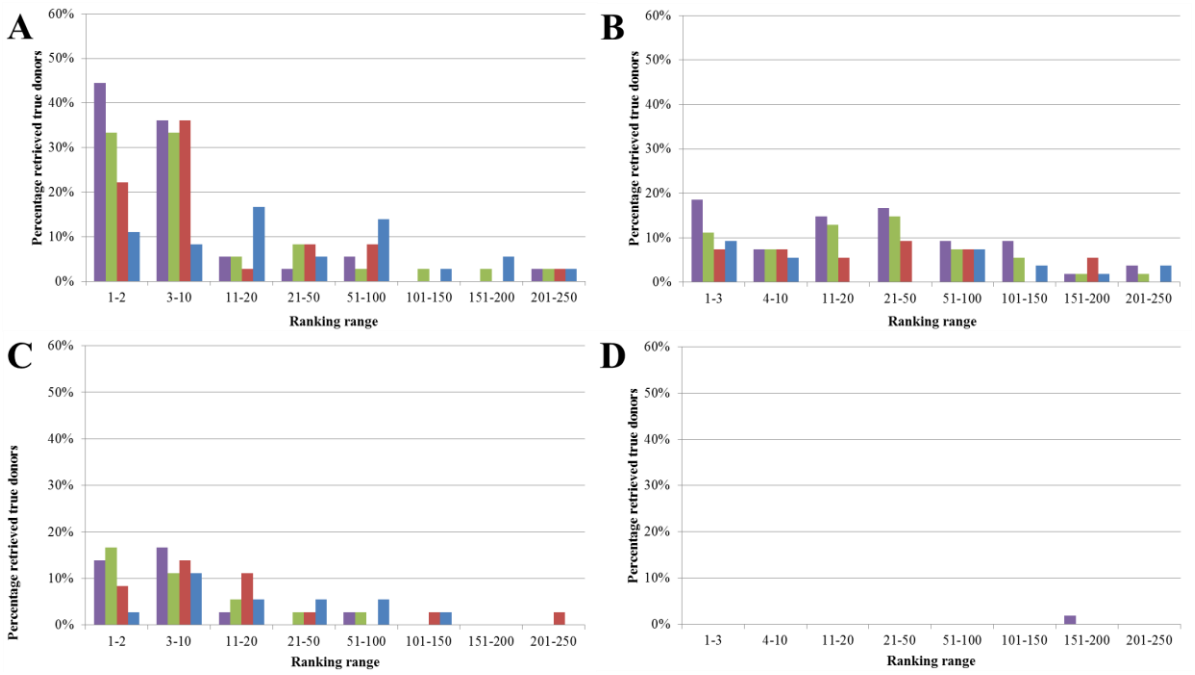


Figure 2. Ranking position for true donors in two-person (A&B) and three-person mixtures (C&D) with 10% (A&C) or 40% drop-out (B&D). The purple, green, red and blue bars represent profiles with eight, seven, six and five loci, respectively. Analyses were performed using the Dutch 2015 DNA database and using nine different loci combinations (Supplementary Table 2).

3.3 SmartRank versus CODIS

Both SmartRank and CODIS are user friendly software enabling DNA database searches. Their potential and purpose, however, is different. CODIS is a powerful software designed to search large numbers of crime scene profiles against a large number of offender profiles daily. CODIS was originally designed to search single donor profiles and up to two-person mixtures without drop-out. SmartRank is designed to be used for DNA database searches using more complex mixed DNA profiles. We investigated for which profile types SmartRank can complement CODIS. To that aim, 30 DNA profiles with various characteristics (both without and with drop-out) were selected from the full list of 343 mixed profiles (Supplementary Table 1). Each of these mixed profiles was used for one SmartRank analysis and two CODIS searches with moderate search stringency: once without allowing mismatches and once while allowing one mismatch. As mentioned previously, we regard a list of ten (possibly 20) candidates as well manageable for further evaluation. Therefore, we also regard whether the true donor ranked within the top ten (or top 20) of the candidate list for SmartRank-based searches and whether the total number of matches did not exceed ten (or 20) candidates when CODIS was used.

Both SmartRank and CODIS retrieved the true contributors successfully for the two-, three- and four-person mixtures without drop-out (Table 4A). Using SmartRank, all true donors were ranked within the top ten. CODIS searches retrieved more than ten candidates when allowing one mismatch

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or when having more contributors. The candidate list can be a truly unmanageable list like for example the 4,302 adventitious matches that occurred for a four-person mixture searched against the Dutch DNA database (Table 4A).

Besides, two- and three-person mixtures with variable level of drop-out and six, eight or 16 loci were used. The occurrence of drop-out and availability of less loci clearly complicates the retrieval of the true donors, especially in CODIS searches that do not allow a mismatch. Although SmartRank did not always retrieve all of the true donors in these searches, true donors (that indeed ranked within the top 10 (or 20) list of candidates) were retrieved more often than with CODIS . With CODIS, large lists of candidate matches occurred especially with few loci or large drop-out levels in the mixtures (Table 4B). SmartRank's potential to complement CODIS is most clear for the 16-loci mixed profiles with low or moderate drop-out (Table 4).

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Table 4. Performance of SmartRank and CODIS for two-, three- and four-person mixtures that (A) lack drop-out or (B) have 5% to 50% overall drop-out. Results for searches using the DNA databases of Belgium, Spain and The Netherlands are combined.

	Number of contributors	Number of loci	Drop-out	Number of samples per DNA database	Total number of searches	SmartRank			CODIS - 0 mismatches		CODIS - 1 mismatch	
						Donors retrieved	Ranking position of retrieved donors		Donors retrieved	Number of adventitious matches	Donors retrieved	Number of adventitious matches
							1-10	1-20				
A	2	15	0%	2	6	100%	100%	100%	100%	2 – 7	100%	3 – 25
	3	15	0%	2	6	100%	100%	100%	100%	3 – 38	100%	5 – 369
	4	15	0%	4	12	100%	100%	100%	100%	4 – 418	100%	7 – 4,302
B	2	16	5%	1	3	100%	100%	100%	0%	-	100%	2 – 4
			15% [a]	1	3	100%	83%	100%	33%	1 – 2	33%	2 – 3
			50%	1	3	0%	-	-	0%	-	0%	0 – 3
		8	10%	2	6	100%	100%	100%	0%	0 – 968	75%	2 – 985
			40%	2	6	100%	100%	100%	0%	0 – 968	83%	1 – 985
		6	10%	2	6	100%	58%	75%	0%	1 – 9	92%	13 – 396
			40%	2	6	100%	25%	100%	0%	1 – 9	100%	13 – 396
	3	16	5%	1	3	100%	100%	100%	22%	0 – 3	44%	4 – 68
			20% [a]	1	3	33%	100%	100%	33%	1 – 2	33%	2 – 4
			50%	1	3	0%	-	-	0%	-	0%	-
		8	10%	2	6	89%	0%	0%	0%	0 – 2,359	89%	10 – 2,461
			40%	2	6	0%	-	-	0%	0 – 317	0%	2 – 326
		6	10%	2	6	67%	0%	0%	0%	0 – 141	56%	64 – 2,399
			40%	2	6	0%	-	-	0%	0 – 10	0%	0 – 154

[a] Major 0% drop-out, minor(s) 30% drop-out.

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3.4 Applicable domain and use of the SmartRank software

When a true donor resides in a DNA database, successful retrieval by SmartRank searches depends both on the complexity of the mixed profile and the composition and size of the DNA database. In this section, and as recommended by the DNA commission of the International Society for Forensic Genetics [13], we infer the scope of application of the software based on the data in this study. In addition, we suggest a LR threshold in combination with a top list of ranked candidates and we examine the drop-out estimation option in the software in order to guide users in obtaining as many true positives and as little false positives as possible.

3.4.1 Specifying a likelihood ratio threshold and a top ranked candidate list

Ideally, true contributors to the DNA mixtures yield LRs >1 and non-contributors yield LRs <1, with a large difference between the two. In order to present an informative list of candidates, we examined the effect of applying a likelihood ratio threshold, *i.e.* a cut-off that excludes the results below this LR value. These analyses are first performed for the top 250 ranking candidates, and subsequently repeated for the top 10 and 20 ranking candidates as this is regarded a manageable number for further investigation. Since the results in the earlier sections showed that samples with high drop-out rates (≥40%) or only a few loci often result in not-informative candidate lists, these sample types are regarded outside the applicable domain of SmartRank and not included in this section that examines the effect of applying a LR threshold.

Results in Table 5 show that the use of a high LR threshold of 100,000 results in few non-donors that exceed this LR, but this comes at costs of missing many of the true donors (LR <100,000) and vice versa (Table 5). None of the tested LR thresholds achieved only the retrieval of the true donors (and no non-donors) when considering all sample types. The most optimal balance between true- and false-positives was obtained when applying a LR threshold of 1,000 or 10,000 (Table 5), although resembling donors may still be retrieved (data not shown). When we applied these LR thresholds of 1,000 and 10,000 to the top 10 and top 20 lists of candidates, much less non-donors were retrieved with little effect on the retrieval of the true donors, especially with the LR>1,000 threshold (results not shown). Thus, we proceeded to use a LR threshold of 1,000 as default value. Since the number of true positives hardly increased when examining the top 20 instead of the top 10 ranking candidates (Table 6), we chose ten candidates for default value. The values for the LR threshold and top list of candidates are, however, configurable in SmartRank and specifically for five-person mixtures with one or two known contributors it may be useful to examine the top 20 ranking candidates to increase the true positive rate. Moreover, a LR threshold of 10,000 may be chosen for less complex mixtures (mixtures without drop-out or two-person mixtures with up to 15% drop-out), as less false positives are expected while still retrieving the true positives (Table 6).

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Table 5. Effect of applying a LR threshold on the effectiveness of retrieving the true donors amongst non-donors for profiles carrying 15 or 16 loci. The candidate lists of the indicated LR threshold and the LR>1 boundary (top 250 ranking candidates only) are compared regarding percentage of true donors missed and percentage of non-donors lost (designed resembling donors excluded). The color scaling presents preferable results in green and less preferable results in orange or red.

Number of contributors	**Percentage of drop-out**	**LR threshold:**	**Number true donors missed** [a]	**Percentage true donors missed**				**Percentage non-donors lost**			
			1	**100**	**1,000**	**10,000**	**100,000**	**100**	**1,000**	**10,000**	**100,000**
2	0%	(*n*=20)	0/40	0%	0%	0%	0%	100%	100%	100%	100%
	5%	(*n*=6)	0/12	0%	0%	0%	0%	97%	98%	99%	100%
	15%	(*n*=5)	0/10	0%	0%	0%	0%	98%	99.5%	100%	100%
	15% [b]	(*n*=5)	0/10	50%	50%	50%	50%	97%	100%	100%	100%
	30%	(*n*=5)	0/10	0%	0%	40%	100%	100%	100%	100%	100%
3	0%	(*n*=20)	0/60	0%	0%	0%	0%	49%	78%	98%	100%
	5%	(*n*=6)	0/18	0%	0%	0%	20%	72%	90%	98%	100%
	15%	(*n*=5)	0/15	0%	20%	33%	67%	78%	97%	99%	100%
	20% [b]	(*n*=5)	10/15	0%	0%	0%	0%	100%	100%	100%	100%
	30%	(*n*=5)	9/15	100%	100%	100%	100%	89%	100%	100%	100%
4	0%	(*n*=20)	0/80	0%	0%	0%	18%	23%	72%	93%	99%
	5%	(*n*=6)	1/18	10%	20%	30%	75%	82%	96%	100%	100%
4 (1 known) [c]	0%	(*n*=20)	0/60	0%	0%	0%	17%	24%	80%	98%	100%
5 (1 known) [c]	5%	(*n*=6)	7/24	7%	13%	53%	87%	60%	95%	99%	100%
5 (2 knowns) [c]	5%	(*n*=6)	4/18	7%	21%	36%	71%	57%	94%	100%	100%

[a] The total number of true donors is the number of (unknown) contributors times the number of SmartRank searches (presented as (*n*=xx)).

[b] Major 0% drop-out, minor(s) 30% drop-out.

[c] Known contributors do not have drop-out and were not searched for.

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Table 6. Percentages of true positives (retrieval of true donors) when applying a LR threshold of 1,000 or 10,000 in combination with examining the top 10 or top 20 list of candidates. SmartRank searches were performed using mixed DNA profiles with 15 or 16 loci.

Number of contributors	**Percentage of drop-out (number of searches)**		**Number of true donors searches for** [a]	**LR threshold:**	**Percentage of true donors retrieved**			
					1,000		**10,000**	
					Top 10	**Top 20**	**Top 10**	**Top 20**
2	0%	(*n*=20)	40		100%	100%	100%	100%
	5%	(*n*=6)	12		100%	100%	100%	100%
	15%	(*n*=5)	10		100%	100%	100%	100%
	15% [b]	(*n*=5)	10		50% [c]	50% [c]	50% [c]	50% [c]
	30%	(*n*=5)	10		100%	100%	60%	60%
3	0%	(*n*=20)	60		100%	100%	100%	100%
	5%	(*n*=6)	18		100%	100%	95%	95%
	15%	(*n*=5)	15		73%	73%	73%	73%
	20% [b]	(*n*=5)	15		33%[c]	33% [c]	33% [c]	33% [c]
	30%	(*n*=5)	15		0%	0%	0%	0%
4	0%	(*n*=20)	80		100%	100%	100%	100%
	5%	(*n*=6)	24		80%	80%	60%	60%
4 (1 known)	0%	(*n*=20)	60		100%	100%	100%	100%
5 (1 known)	5%	(*n*=6)	24		55%	65%	35%	35%
5 (2 knowns)	5%	(*n*=6)	18		56%	61%	50%	50%

[a] Number of 'unknown' contributors times the number of SmartRank searches (presented as (*n*=xx)).
[b] The Major contributor has 0% drop-out, the minor(s) 30% drop-out.
[c] For each of these mixtures the major contributor was retrieved, while the minor(s) were not.

3.4.2 The effect of the probability of drop-out parameter on the retrieval and ranking of true donors

In SmartRank, the probability of drop-out can either be set manually or it can be estimated by the software following a MCMC method. In order to investigate the effect of the probability of drop-out on the retrieval and ranking of the true donors to the DNA mixtures, we compared results when using the known ('true') drop-out value, the 5%, 50% and 95% percentile of the drop-out estimation for a selection of 25 mixed DNA profiles (Supplementary Table 1). Best results were obtained when using the known drop-out rate (Table 7), though this rate is never known in casework and can be challenging to infer. When using the drop-out estimation option and considering all types of mixtures, the highest percentages of true positives ranking in the top ten candidate list with a LR >1,000 were obtained using the 50% percentile. When regarding the mixture types specifically, the two-person mixtures showed that also the 95% percentile, which is the default value that is currently set in SmartRank, could be used as it showed similar results. As it may be desired to perform a search using another percentile, this value will be made configurable in the SmartRank software version 1.0.3 and onwards.

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Table 7. Percentages of true donors that were ranked in the top 10 list of candidates having a LR >1,000 when using the known drop-out rate or the 5%, 50% or 95% percentile of the drop-out estimation in SmartRank.

Number of contributors	Percentage of drop-out (number of samples)		Number of true contributors in total	Percentage of true positives in top 10 and with LR >1,000			
				Known drop-out applied	Drop-out estimation in SmartRank		
					0.05 quantile	0.5 quantile	0.95 quantile
2	0%	(*n*=4)	8	100%	100%	100%	100%
	5%	(*n*=2)	4	100%	50%	100%	100%
	15%	(*n*=1)	2	100%	100%	100%	100%
	30%	(*n*=1)	2	100%	100%	100%	100%
3	0%	(*n*=4)	12	100%	100%	92%	50%
	5%	(*n*=2)	6	83%	67%	83%	83%
	15%	(*n*=1)	3	67%	67%	67%	0%
4	0%	(*n*=4)	16	100%	56%[a]	50%[a]	50%[a]
	5%	(*n*=6)	24	79%	29%	71%	46%
	Total	(*n*=25)	77	91%	62%	77%	60%

[a] Two of the four-person mixtures without drop-out had low allele sharing and two had high allele sharing between the contributors. Contributors were not retrieved for the high allele sharing mixtures as the drop-out rate for these mixtures were overestimated (by more than 20%).

4. **Concluding remarks**

SmartRank was developed to put forward only relevant candidates from a DNA database search and rank them accordingly. To this aim, the SmartRank software computes a LR for the searched profile and each profile in the DNA database and ranks database entries above a defined LR threshold according to the calculated LR. We validated SmartRank using many mixed DNA profiles and databases of various compositions. In order to assist users of the SmartRank software, we provide guidelines for best practice that follow from the data described in this validation study. These are not to be confused with standards. Note that to enable flexibility, several options in SmartRank are made configurable by the user. For instance, we decided to set the default value for the top list of candidates to 10 and the LR threshold to 1,000 as this has shown to minimize false negatives and false positives with the data and DNA databases used in this study, however these values can be changed by the users. A guide for deciding whether a SmartRank DNA database search is expected to be successful concerning the combination of the number of contributors and percentage of drop-out is presented in Table 8. This guide is based on STR profiles containing 15 or 16 loci and a DNA database of size 37,000 - 1.55 million. Searching with profiles containing less than 15 loci is likely to negatively influence the retrieval of true donors. The presence of a rare allele, on the other hand, can have a positive effect on the retrieval of true donors. Searching with profiles containing larger numbers of loci (*e.g.* 23 loci) is expected to extend the applicable domain of the software; more complex DNA profiles may be suitable for a SmartRank search. Using a DNA database that is larger in size than

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tested in this study can result in larger numbers of false positives, and smaller DNA databases is expected to yield less false positives.

Depending on the type of the case and keeping in mind the expected trends, the scientist can decide to perform a SmartRank search for DNA mixtures with characteristics in regards to the orange boxes in Table 8. The user may wish to extend the list of candidates to, for example, 20 as this can result in more true positives for some of the mixtures, though the number of false positives may also increase.

Overall, whether or not a SmartRank DNA database search is successful depends on the combinations of the size and composition of the DNA database and the various mixtures characteristics, and, of course, whether or not the true contributors to the DNA mixtures are present in the DNA database.

Table 8. Guide for deciding whether a SmartRank DNA-database search can be successful for the type of DNA-profile at hand. This guide follows from the data described in this study. Expectations are for STR profiles containing 15 or 16 loci and using a DNA database of size 37,000-1.55 million.

Percentage of drop-out	**Number of contributors**				
	2	**3**	**4**	**5 (1 known)**	**5 (2 knowns)**
0% [a]	Successful SmartRank search expected				
5% [b]			The loss of true donors is expected		
15%		Successfulness dependent on complexity; further considerations advised			
30%		Some or (all) donors may be missed			
50%	SmartRank searches did not yield informative results and are therefore not advised				

[a] For two-person mixtures without drop-out a CODIS search may be successful too.

[b] For five-person mixtures with 5% drop-out and known contributors, extending the maximum list of candidates to 20 is expected to yield more true positives.

In summary, SmartRank is an open-source and user-friendly software that enables searching voluminous DNA databases with complex DNA profiles. The applicable domain of the software was defined and using the provided guidelines for best practice, SmartRank enables obtaining investigative leads in (criminal) cases lacking a suspect. The SmartRank software is freely available for download through http://lrmixstudio.org/smartrank.

Acknowledgements

This study received funding support from the European Network of Forensic Science Institutes (ENFSI) Monopoly programme "SmartRank: A likelihood ratio software for searching national DNA databases with complex DNA profiles" (MP2013-T4). We are thankful to professor Peter Gill and Lourdes Prieto Solla for valuable discussions regarding SmartRank design. Lieutenant

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Colonel CC Andrea Berti from the Forensic Biology Unit of the RIS of Rome for his valuable scientific support and suggestions in statistical analyses, and Warrant Officer CC Daniele Mondelli from the Logistic Unit of the RaCIS for the ICT support in CCDNA DNA Database data management. Sergio Hernanz Asensio and César del Caño Sánchez, are thanked for assistance with the Spanish database searches in CODIS. Adrián Rodríguez Menchén and Marc Laman Trip are thanked for making the copy of, respectively, the Spanish and Netherlands database and managing the computer processes for the good end of the exercises. Mariane Linger, Manorma Autar, Charissa van Kooten and Jord Nagel are thanked for their assistance with the SmartRank validation at the Netherlands Forensic Institute.

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